

Exact Radial-Determinant Flexural Frequencies of a Monolithic Thickness-Poled Piezoelectric Ring: An Open-Circuit-Equals-Short-Circuit Theorem and a Segmented-Electrode Sensing Admittance

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Abstract

A homogeneous, thickness-poled piezoceramic annulus – one ceramic spanning the full plate thickness, electroded only on its two faces, with no elastic host – is solved by the same exact radial-determinant method used for a layered piezoelectric sandwich, but with a different through-thickness potential. Two results follow from removing the offset skins. First, because a spatially uniform electrode voltage produces a field symmetric about this plate’s own neutral surface, its piezoelectric moment integrates to zero for every mechanical boundary condition tested: fully electroded open-circuit flexure coincides exactly with short-circuit flexure, and the electrode-voltage admittance degenerates to a frequency-independent capacitor, $Y = j\omega C_0$ with $k_{\text{eff}}^2 \equiv 0$. Second, the short-circuit frequency itself becomes the identifying piezoelectric signal, sitting +9.6% (free-clamped) to +13.8% (clamped-free) above the same plate’s purely elastic frequency on a representative PZT-4 geometry – several times the open-circuit split of a steel-dominated layered sandwich of the same material, and well conditioned for inverting the piezoelectric stress constant e_{31} : a 0.01% frequency uncertainty propagates to only 0.15%–0.17% of e_{31} . The through-thickness potential is closed by a Galerkin projection of the electric enthalpy, which makes the long-wave short-circuit stiffening exact; the thickness-integrated Gauss law used for layered plates overstates it by $12/\pi^2$. The elastic limit agrees with an independent closed-ring solver to 4.8×10^{-10} relative, and finite elements confirm both the coincidence and the stiffening magnitude, reproducing the closed-form split to 0.03% of its value. Because the voltage-driven route is trivial, sensing instead reads the mechanical side of the same coupling: a harmonic radial force drives a genuinely resonant charge response on a segmented electrode, even though the same force drives an identically zero charge on the full electroded face. That segment charge carries a transverse-shear piezoelectric (e_{15}) contribution that Kirchhoff kinematics discard but that enters at the same order as the dielectric flux. With it, the charge is a closed-form constant multiple κ of the dielectric-flux value, $\kappa = -1.96$ for PZT-4, which coupled-field finite elements confirm at the tabulated shear stiffness to within 0.5% on the free-free ring and within 4% on clamped and mixed edges, provided the partition sits well clear of a clamped rim. A symmetric, thickness-poled monolithic plate can therefore be neither voltage-actuated nor full-electrode-sensed in flexure, but can be force-driven and segment-sensed – the textbook justification for unimorph and bimorph bending actuators, derived here as an exact, closed-form property of the radial-determinant operator.

Keywords: piezoelectric annular plate; monolithic piezoceramic; open-circuit; short-circuit; electromechanical coupling; radial determinant; free-free flexure; sensing admittance.

1 Introduction

A companion paper from this group solved a layered annular plate – an isotropic host with identical thickness-polarized piezoceramic skins bonded to both faces – by an exact radial-determinant method, and used the resulting open-circuit flexural frequency to invert the piezoelectric stress constant e_{31} at a precision the corresponding short-circuit frequency could not reach [10]. That construction is a deliberate choice of specimen: a thin piezoceramic layer offset from the plate’s neutral axis, so that even a spatially uniform electrode voltage produces a bending moment. It is not, however, the specimen most readily available for a physical test. A commercial thickness-poled piezoceramic ring – the kind sold as a ultrasonic transducer element or a buzzer disk – is not a host with bonded skins; it is one ceramic, poled through its full thickness, electroded on its two faces. The present paper asks the same questions of that geometry.

The answers are qualitatively different, not merely quantitatively so. On the layered plate, the piezoceramic skins sit at $|z| > h$, offset from the neutral surface, so a spatially uniform through-thickness field has a nonzero moment arm and free–free open-circuit flexure genuinely splits from short-circuit flexure. On the homogeneous monolithic plate the ceramic *is* the neutral surface: a spatially uniform field is symmetric about it, and the piezoelectric moment it would produce integrates to zero over $|z| \leq H$ for every mechanical boundary condition, not only the clamped–clamped case where the layered plate’s own coincidence already held. That is the textbook reason real bending-mode piezoelectric actuators are built as unimorphs or bimorphs and never as symmetric monolithic slabs: this paper derives that fact as a closed-form theorem for the exact radial-determinant operator, rather than assuming it, and then asks what *is* identifiable and sensible on the specimen that is actually a single poled ceramic.

Two results follow directly from removing the offset skins. First, the short-circuit flexural frequency itself – not an open-circuit split – becomes the piezoelectric signal, because the whole plate thickness is now piezoceramic rather than a thin fraction of it: the short-circuit coupling on this stacking is about ten percent in frequency, where it was two to three orders of magnitude smaller on the layered plate. Second, because the voltage-driven admittance degenerates to a frequency-independent capacitor, the free–free resonance/antiresonance identification used on the layered plate is not available here. Sensing instead uses the mechanical side of the same electromechanical coupling: a harmonic radial force on the plate drives a genuinely resonant charge response, provided the sensing electrode is a proper subset of the grounded face rather than the whole face – the full-face charge is itself identically zero, by the same cancellation that kills the voltage-driven admittance, read from the other electrical port.

Parashar gave a Rayleigh–Ritz treatment, with experiment, of a radially poled annular piezoceramic actuator whose flexural actuation mechanism is shear-induced (d_{15}) rather than thickness-poled bending, on Mindlin rather than Kirchhoff kinematics [8]. That is a different poling direction, a different actuation mechanism, and a different plate theory; the coincidence theorem of this paper does not apply to it, and it is not scored here. It is a natural second phase on the same family of exact radial-determinant methods and is left for later work. Holland’s contour-mode analysis of an unpoled elastic disk [6] is the classical reference for the in-plane (extensional, k_p) response the same physical ring would show on its impedance spectrum; that response is disclosed here as a boundary of scope, not built.

The underlying elastic operator, and the exact radial-determinant method itself, are the closed-annulus reduction used throughout this project’s series [9]: integer circumferential order n , a boundary-condition-selected radial edge matrix, and frequencies as zeros of that determinant. The layered-plate piezoelectric extension [10] added a through-thickness potential that vanishes on every electrode, after Duan, Quek and Wang [3], and the constitutive reduction of Liu, Wang and

Quek [4]; the present paper reuses that constitutive reduction on a single ceramic spanning the full thickness, with a potential shape that is even, not odd, about the plate's own midplane, which is the root cause of every difference from the layered result. Both papers close the electrical equation by the same Galerkin projection of the electric enthalpy that produces the mechanical one, rather than by Duan, Quek and Wang's thickness-integrated Gauss law; §2.2 explains why that choice changes the short-circuit stiffening by a fixed factor.

What is not claimed: in-plane (extensional, contour-mode) piezoelectric coupling as a headline result, though a negative control confirming that a uniform voltage *does* couple to membrane force is included; a radially poled or shear-actuated (d_{15}) specimen; a laboratory specimen; or a joint fit of elastic stiffness and e_{31} from a single frequency.

2 Theory

2.1 Kinematics and constitutive reduction

The ring occupies $r_i \leq r \leq r_o$, $|z| \leq H$ (full thickness $2H$), one homogeneous piezoceramic, thickness-polarized, electroded only on the two faces $z = \pm H$. There is no elastic host and no electrode at the midplane. Kirchhoff kinematics are unchanged from the elastic and layered-piezoelectric cases,

$$u_r = -z \partial_r w, \quad u_\theta = -z r^{-1} \partial_\theta w, \quad u_z = w(r, \theta, t), \quad (1)$$

giving strains odd in z and $\varepsilon_{rr} + \varepsilon_{\theta\theta} = -z \nabla^2 w$. Transverse shear strain vanishes identically in these kinematics, so e_{15} does not enter any frequency in this paper, as in the layered case. It does enter the segment charge of §5.2 at leading order, through the transverse shear stress that equilibrium requires.

Plane-stress reduction of the full three-dimensional piezoceramic constants follows the same construction as the layered plate [4, 10]:

$$\bar{c}_{11}^E = c_{11}^E - \frac{(c_{13}^E)^2}{c_{33}^E}, \quad \bar{e}_{31} = e_{31} - \frac{c_{13}^E}{c_{33}^E} e_{33}, \quad \bar{\Xi}_{33} = \Xi_{33} + \frac{e_{33}^2}{c_{33}^E}. \quad (2)$$

At the material point used throughout this paper (§4), $\bar{e}_{31} \approx -13.05 \text{ C m}^{-2}$: the correction $(c_{13}^E/c_{33}^E)e_{33} = 8.95 \text{ C m}^{-2}$ has the same sign as e_{31} and dominates it, the same reduction as on the layered plate. Bending stiffness has no separate host or layer terms; the whole thickness is piezoceramic,

$$d = \frac{2}{3} \bar{c}_{11}^E H^3, \quad \nu_p = \bar{c}_{12}^E / \bar{c}_{11}^E, \quad A_1 = \frac{1}{2} (1 - \nu_p) d, \quad A_2 = 2 \rho H. \quad (3)$$

The piezoelectric extra stress in σ_{rr} and $\sigma_{\theta\theta}$ is $-\bar{e}_{31} E_z$. Its contribution to the membrane resultant $N_{rr} = \int_{-H}^H \sigma_{rr} dz$ and to the bending resultant $M_{rr} = \int_{-H}^H z \sigma_{rr} dz$ splits on the parity of E_z about the plate midplane: an E_z that is even in z (a spatially constant field, produced by a uniform electrode voltage) contributes to N_{rr} and integrates to exactly zero in M_{rr} , because z is odd and the interval $[-H, H]$ is symmetric; an E_z that is odd in z does the reverse. On the layered plate the piezoceramic sits entirely at $|z| > h$, so even a spatially constant field has a nonzero moment arm about the neutral axis and produces a genuine bending moment. On this monolithic plate the ceramic occupies the neutral surface itself, so the even/constant field is symmetric about it and contributes no moment at all, for any thickness or any material constants. That single parity fact is the source of every result in this paper: a spatially uniform face voltage is, on this stacking, an extensional actuator and not a flexural one.

2.2 Short-circuit potential and the closing cubic

With both faces grounded, $\phi(\pm H) = 0$, the potential ansatz is an even quadratic that vanishes at both faces,

$$\phi(r, \theta, z, t) = \bar{\phi}(r, \theta, t) \left(1 - \frac{z^2}{H^2}\right). \quad (4)$$

ϕ is even in z ; the interior field $E_z = 2z\bar{\phi}/H^2$ is odd, and is therefore the flexural actuator identified in §2.1. The equation for $\bar{\phi}$ comes from the same projection as the flexural equation: stationarity of the thickness-integrated electric enthalpy with respect to $\bar{\phi}$, with the ansatz itself as the weight. The alternative used for layered plates by Duan, Quek and Wang [3] closes the electrical equation by integrating Gauss's law through the thickness, i.e. with a unit weight, while the coupling reaches the flexural equation through the moment, i.e. with weight z . The two projections are then not conjugate, and the long-wave short-circuit stiffening depends on which is used. With a sinusoidal profile, $\bar{\phi} \cos(\pi z/2H)$, the Gauss-law closure overstates the exact long-wave increment by exactly $12/\pi^2 = 1.216$, and the same-weight projection understates it by $96/\pi^4 = 0.986$ (Supplementary § S.1). The quadratic (4) is the unique one-term profile for which the same-weight projection is exact in the long-wave limit: it reproduces the thin-plate short-circuit result obtained directly from $\partial_z D_z = 0$ with $\phi(\pm H) = 0$, which is $\bar{e}_{31}^2/\bar{\epsilon}_{33}$ added to both $\bar{c}_{11}^E$ and $\bar{c}_{12}^E$. Substituting into the moment/shear resultants and the projected electrical equation, and positing $\bar{\phi} = \chi \bar{w}$ with χ independent of r , again produces two Helmholtz equations for $\bar{w}$ that are simultaneously satisfiable only when χ and the radial eigenvalue λ satisfy a cubic consistency condition in χ (Supplementary § S.1); the coefficients of that cubic are not obtained by substituting the layered plate's own layer thickness $h_1 \rightarrow 2H$ into its cubic, because the two potential shapes have different parity about the plate midplane. The cubic has three roots (χ_i, λ_i) , each a function of the trial frequency ω ; for each root the radial problem is second-order Helmholtz, with Bessel pair (J_n, Y_n) or modified pair (I_n, K_n) by the sign of λ_i . Three branches times two radial functions is a six-dimensional basis, matching the sixth-order coupled system, exactly as on the layered plate. Setting $\bar{e}_{31} \rightarrow 0$ is again a singular limit of the cubic, so the purely elastic ring is assembled as a separate two-branch 4×4 , gated independently in §2.4.

Edge conditions are mechanical plus the insulated-rim electrical condition $\partial \bar{\phi} / \partial r = 0$ at r_i and r_o . Four mechanical edge pairings are used in this paper, in Paper 3's inner-then-outer naming [9]: free-free ($M_{rr} = Q_r^{\text{eff}} = 0$ at both edges, with $Q_r^{\text{eff}} = Q_r + r^{-1} \partial M_{r\theta} / \partial \theta$ the Kirchhoff effective shear), clamped-clamped ($w = w' = 0$ at both edges), clamped-free (inner clamped, outer free), and free-clamped (the swap). Each pairing selects the corresponding rows of the 6×6 short-circuit determinant, with the two insulated-rim rows unchanged, and the free-free and clamped-clamped determinants are literal special cases of the same mixed-edge row-selection logic used for the clamped-free and free-clamped pairs (Supplementary § S.6). Zeros are found by a real-part sign-flip bisection on a pre-registered frequency bracket, with a hi-side-only negative-control bracket that is not permitted to flip; published values, where any exist, are never used to seed a search.

2.3 Open-circuit flexure coincides with short-circuit flexure

Open-circuit replaces the grounded faces with a single floating bus at unknown voltage $V(t)$, with total electrode charge constrained to zero. The leading through-thickness potential is the linear particular solution plus the same interior profile,

$$\phi = \frac{V(t)z}{2H} + \bar{\phi}(r, \theta, t) \left(1 - \frac{z^2}{H^2}\right). \quad (5)$$

The linear term's field $E_z^{\text{lin}} = -V/(2H)$ is spatially constant – even in z – so, by §2.1, it contributes identically zero to M_{rr} for any V , any geometry, any material constants. It also does not enter Q_r^{eff} or the interior field equations: the bus is a single equipotential surface with no r, θ structure, so its field has zero radial gradient, and its cross term with the interior profile integrates to zero through the thickness ($\int_{-H}^H \partial_z(1 - z^2/H^2) dz = 0$). The projected electrical equation and the flexural motion equation are therefore exactly the short-circuit pair, driven by $\bar{\phi}$ alone. The unknown bus voltage V is not a second frequency-domain unknown; it is fixed *after* the mode is known, by the single scalar constraint $Q = 0$. The bus charge conjugate to V in the same enthalpy has no motional part at all – the linear field's strain coupling carries the odd weight $\int_{-H}^H z dz = 0$ and its cross term with $\bar{\phi}$ vanishes as above – so

$$Q = -C_0 V, \quad C_0 = \frac{\bar{\Xi}_{33} A}{2H}, \quad A = \pi(r_o^2 - r_i^2), \quad (6)$$

and $Q = 0$ gives $V = 0$: a flexural mode develops no voltage on an open-circuit full-face bus. **Theorem.** For a homogeneous, thickness-polarized, fully electroded Kirchhoff annulus under this ansatz, open-circuit flexural frequencies coincide with short-circuit flexural frequencies exactly, for every mechanical boundary condition. The IEEE electromechanical coupling factor $k_{\text{eff}}^2 = 1 - (f_r/f_a)^2$ built from flexural resonance and antiresonance is identically zero. A spatially uniform electrode voltage cannot actuate flexure on this specimen.

This is the reverse of the layered plate's own result. There, offset skins give a spatially constant field a nonzero moment arm, so free–free open-circuit genuinely splits from short-circuit and only the clamped–clamped pairing coincides, because Green's identity forces the electrode-charge term that carries V to vanish when the edge rotation is zero everywhere [10]. Here the coincidence holds regardless of the mechanical edge, because V never enters the moment balance at all – the geometric offset that made the layered plate's coincidence a special case of one boundary condition does not exist on a symmetric monolithic plate. Do not invert an open-circuit-versus-short-circuit split for e_{31} on this stacking, and do not add a frequency solver expecting one; by (6) the full-face electrode does not even sense the flexural mode, which is why §5.2 partitions it.

2.4 Elastic limit

Setting $\bar{e}_{31} = 0$ (and $\bar{\phi} = 0$) recovers an isotropic Kirchhoff ring of thickness $2H$ with bending stiffness $D = d$ of (3) and Poisson ratio $\nu = \nu_p$. Matching the standard isotropic-plate stiffness $D = Eh^3/[12(1 - \nu^2)]$ at $h = 2H$ requires

$$E = \bar{c}_{11}^E (1 - \nu_p^2) = \frac{(\bar{c}_{11}^E)^2 - (\bar{c}_{12}^E)^2}{\bar{c}_{11}^E}, \quad \nu = \nu_p = \bar{c}_{12}^E / \bar{c}_{11}^E. \quad (7)$$

This is the first numerical gate applied in §4, before any coupled number is trusted: the purely elastic 4×4 at $(E, \nu, \rho, h = 2H)$ from (7) must agree with an independently implemented closed-annulus Kirchhoff solver [9] that shares no code with the piezoelectric construction.

Long-wave reduction of the coupled system (dropping $\Delta\bar{\phi}$ against $\bar{\phi}$ in the projected electrical equation) gives a closed-form stiffening estimate,

$$D_{\text{SC}} = d \left(1 + \frac{\bar{e}_{31}^2}{\bar{\Xi}_{33} \bar{c}_{11}^E} \right) > d, \quad (8)$$

which is the exact thin-plate short-circuit bending stiffness; the Gauss-law closure would put $12/\pi^2$ in front of the correction. The estimate is independent of the sign of $\bar{e}_{31}$: short-circuit flexure sits

above the purely elastic frequency on this stacking, for any material point, and because open-circuit coincides with short-circuit there is no further open-circuit-specific stiffening at this order. This is the sign gate checked against every numerical branch in §4.

3 Numerical method

Frequencies are zeros of a boundary determinant assembled from the Helmholtz/Bessel radial functions of §2.2–§2.4, using the same sign-flip bisection, worker-process reconstruction from scalar material and geometric constants, and row/column equilibration discipline as the elastic and layered-piezoelectric solvers in this series [9, 10]; every root reported below was independently reproduced through a standard multiprocess worker-reconstruction path, not only in-process, before being trusted. The free–free and clamped–clamped determinants and the clamped–free/ free–clamped pair are built from one shared mixed-edge row-selection routine, so that the two originally implemented boundary conditions and the two added later are literally the same code exercised with different edge arguments; this is verified bit-for-bit in Supplementary § S.6, not only by a fresh derivation, which is a stronger correctness guarantee than an independent re-derivation would give, because any defect in the row-selection logic would almost certainly have broken the already-validated free–free and clamped–clamped paths as well.

A second, driven problem is built on the same short-circuit operator for the sensing admittance of §5: a harmonic radial ring force at $r = r_F$ splits the domain into two radial regions, each carrying the same three Helmholtz branches as the free-vibration problem, twelve constants in total. Integrating the loaded flexural balance across the load radius shows that only the Kirchhoff shear Q_r jumps there; the deflection, slope, bending moment, potential, and potential slope are all continuous, because none of the governing equations admits a delta-function source in those quantities. Six mechanical and electrical rim conditions (three per edge) plus six interface conditions – five continuity statements and the one inhomogeneous shear jump – close a twelve-unknown linear system at every trial frequency, which is solved rather than searched for zeros; away from a resonance the system is well conditioned once row and column equilibration is applied, which is necessary here because the third (evanescent) Helmholtz branch produces radial arguments large enough that raw matrix entries span many orders of magnitude. The homogeneous version of this same twelve-unknown system – no applied force – has a nontrivial kernel exactly when the two-region fields glue into a single global mode, by uniqueness of continuation for the underlying sixth-order radial operator; consequently its resonances sit at the already-validated free-vibration roots of §2.2, which is verified directly rather than assumed (§5).

4 Forward validation

Numerical results use PZT-4 constants transcribed from Duan, Quek and Wang’s Table 1 [3], the same source and the same values used for the layered plate [10], so that the two papers compare the same material at two stackings rather than two materials: $c_{11}^E = 132$, $c_{12}^E = 71$, $c_{13}^E = 73$, $c_{33}^E = 115$ GPa, $e_{31} = -4.1$, $e_{33} = 14.1$ C m^{−2}, $\Xi_{11} = 7.124 \times 10^{-9}$, $\Xi_{33} = 5.841 \times 10^{-9}$ F m^{−1}, $\rho = 7500$ kg m^{−3}, including the sign of e_{31} as printed there and in Liu, Wang and Quek’s Table 1 [4]. Every frequency effect below scales with $\bar{e}_{31}^2$; the companion paper shows that Duan, Quek and Wang’s own Table 4 and finite elements both resolve the printed sign [10]. The segment charge of §5.2, and nothing else in this paper, also uses $e_{15} = 10.5$ C m^{−2} and the transverse shear stiffness $c_{44}^E = 26$ GPa of Liu, Wang and Quek [4]. Duan, Quek and Wang print 73 GPa for this constant, which we take to be their c_{13}^E repeated, since PZT-4 is near 26 GPa. Geometry is $r_i = 0.1$ m,

$r_o = 0.6$ m, full ceramic thickness $2H = 0.02$ m ($r_o/H = 60$), the same radii as the layered plate's host, with the steel replaced entirely by ceramic. Figure 1 is the through-thickness geometry.

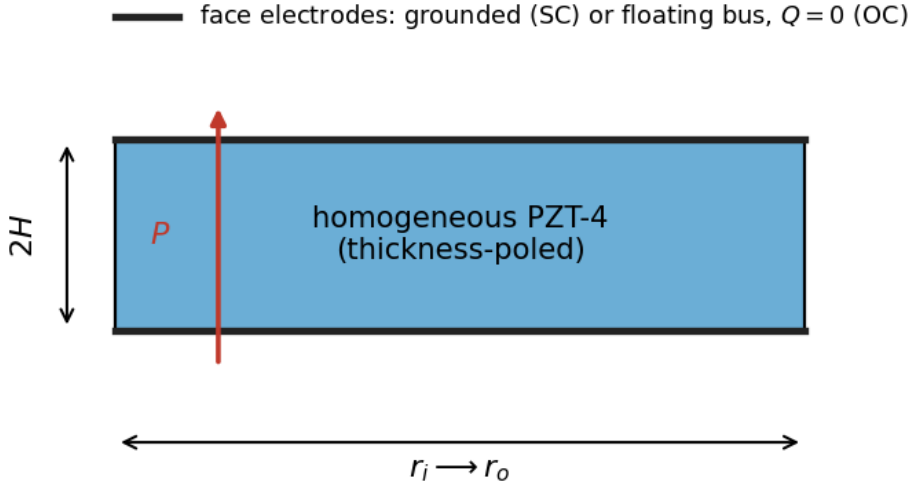


Figure 1: Radial–thickness slice of the monolithic ring (not to scale). One homogeneous, thickness-poled piezoceramic spans the full plate thickness $2H$; both faces are electroded, grounded under short-circuit or forming a single floating bus under open-circuit.

4.1 Elastic-limit gate

At $(E, \nu) = (78.561 \text{ GPa}, 0.28789)$ from (7), the purely elastic free–free $n = 0$ fundamental of the new mixed-edge 4×4 agrees with an independently implemented closed-annulus Kirchhoff solver [9] to 4.8×10^{-10} relative (Supplementary § S.2). This is the gate that must pass before any coupled number below is trusted, and it uses none of the piezoelectric code path.

4.2 Short-circuit stiffening across four mechanical boundary conditions

Table 1 reports the first $n = 0$ flexural root of the purely elastic ring and of the coupled short-circuit 6×6 (which, by the theorem of §2.3, is simultaneously the open-circuit root), on all four mechanical edge pairings. Short-circuit sits above elastic at every pairing, as required by the sign of (8), and the split is about ten percent rather than the 0.005–0.05% found on the layered plate's own short-circuit channel [10]: with no elastic host to carry most of the bending stiffness, the piezoelectric coupling is no longer a small correction to an otherwise elastic plate. The clamped–free and free–clamped rows show that the split is not a property of the free–free geometry alone, and that it is sensitive to which edge is free: at this radius ratio the inner edge dominates bending stiffness far more than the outer edge does (the clamped–free elastic fundamental sits 90.4% of the way from the free–free fundamental to the clamped–clamped fundamental, while free–clamped sits only 7.5% of the way there), a finding specific to this r_i/r_o and reported as such, not as a universal ranking of edges. Figure 2 shows the same four splits as a bar chart.

Table 1: First $n = 0$ flexural frequency, purely elastic ring versus the coupled short-circuit operator (which coincides with open-circuit, §2.3), all four mechanical edge pairings, in rad s^{-1} (Hz in parentheses). C–F is clamped inner, free outer; F–C is the swap.

edge	elastic	SC = OC	split
F–F	462.398 (73.593)	519.839 (82.735)	+12.422%
C–C	1726.890 (274.843)	1938.861 (308.579)	+12.275%
C–F	1605.443 (255.514)	1827.039 (290.782)	+13.803%
F–C	556.884 (88.631)	610.530 (97.169)	+9.633%

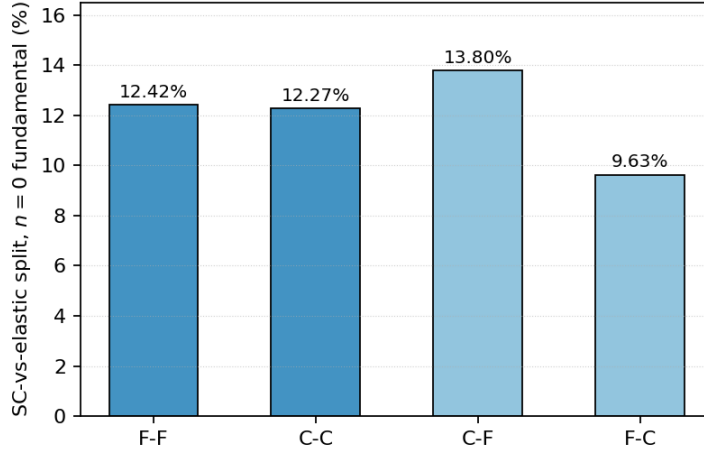


Figure 2: Short-circuit-versus-elastic split of the $n = 0$ fundamental, Table 1, by mechanical boundary condition.

The stiffening persists beyond the fundamental and beyond one thickness. A sign-flip scan of the free–free channel up to 6000 rad s^{-1} finds two further overtone pairs, both still short-circuit above elastic (+10.55% and +11.01%, against +12.42% at the fundamental – the split is not monotonic in overtone index); the clamped–clamped channel stays flatter over one further overtone (+12.18%, against +12.27% at the fundamental). A nine-point sweep of the ceramic thickness from $H/r_o = 0.005$ to 0.18, at fixed radii, confirms a falsifiable prediction of the long-wave reduction of §2.4 – that the split is independent of H at leading order – below $H/r_o \approx 0.025$, where the split stays flat near 12.42%; above that it drifts downward by about 4% relative by $H/r_o = 0.18$, a finite-thickness correction beyond the long-wave leading order, consistent with Kirchhoff kinematics being best justified in exactly the thin regime this paper’s own geometry sits in ($H/r_o \approx 0.017$). Full numeric tables for both scans are in Supplementary § S.5.

4.3 Finite-element cross-check

An independent coupled-field finite-element model (axisymmetric, piezoelectric-plane elements, Lanczos eigensolver), built on the same mesh, material block, and electrical-boundary conventions already validated for the layered plate [10], with the host volume deleted and the piezoceramic volume filling the full thickness, confirms both the coincidence theorem and the stiffening magnitude on free–free $n = 0$. The finite-element open-circuit and short-circuit flexural modes agree with each other to all eight printed significant figures – the coincidence theorem is not only a Kirchhoff-theory artifact, since the finite-element model carries genuinely two-dimensional coupled-field kinematics –

and both sit 12.426% above an uncoupled elastic control on the same mesh, against the closed-form +12.422% of Table 1: the finite-element split reproduces the closed-form split to 0.03% of its value. Run with $e_{31} = +4.1$, the same decks give +1.942% against a closed-form +1.939%, so the finite elements follow the $\bar{e}_{31}^2$ dependence, not one calibrated point. The +4.1 comparison is also what exposed the projection issue of §2.2. The Gauss-law closure predicts +2.347% there, of which finite elements recover only 83%, and $1/0.83 \approx 12/\pi^2$; the same $12/\pi^2$ ratio separates that closure from a three-dimensional finite-element model of a clamped piezoceramic annular sector (Supplementary § S.3). No spurious zero-frequency mode is present other than the expected free-free rigid body, and the bending mode is identified by the same through-thickness displacement discriminator used throughout this series: transverse displacement of the same sign, in-plane displacement of opposite sign, at the two faces at the outer radius. A companion extensional mode on the same mesh does split between open- and short-circuit (+10.95%) – confirming, as a disclosed side observation and not a Paper 5 result, that the linear (membrane) field does couple to extension even though it does not couple to bending, the negative control anticipated in §2.1. Full provenance is in Supplementary § S.3.

4.4 e_{31} inverse from short-circuit-versus-elastic frequency

With $\bar{c}_{11}^E$ (equivalently E, ν through (7)) held at its known value, e_{31} is the sole unknown, recovered by the same residual-bisection and on-branch sensitivity recipe used throughout this series [9, 10]: a target frequency at the true e_{31} is bisected for, on a bracket containing a genuine sign flip, with a wrong-sign bracket and an $e_{31} = 0$ singular-limit check as negative controls. Round-trip recovery of the injected e_{31} from its own forward frequency is exact to 2.5×10^{-10} relative (bar 10^{-6}) on both the free-free and clamped-clamped channels; the inverse machinery is not the issue, and the scientific content is the propagated uncertainty. On-branch sensitivity gives $d\omega/de_{31} = -7.661 \text{ rad s}^{-1} \text{ per C m}^{-2}$ (free-free) and -30.708 (clamped-clamped); propagating a 0.01% relative frequency uncertainty through those sensitivities gives 0.17% and 0.15% of e_{31} , respectively, two to three orders of magnitude better conditioned than the short-circuit inverse reported for the layered plate (33–330% over its three thickness ratios) [10], and 2.6 to 5.6 times better than that paper’s own *open*-circuit inverse (0.45–0.87%) – because on this stacking the short-circuit frequency already carries the full interior-profile coupling that, on the layered plate, only the open-circuit channel could access. Do not invert an open-circuit-versus-short-circuit split for e_{31} here: no such split exists (§2.3), and this single-observable inverse does not jointly fit $\bar{c}_{11}^E$ and e_{31} from one frequency.

One structural caveat applies to this inverse and is proved, not only observed: the closing cubic of §2.2 is invariant, as a set of determinant rows up to an overall rescaling that does not move its zeros, under the reflection $\bar{e}_{31} \rightarrow -\bar{e}_{31}$ composed with $\chi \rightarrow -\chi$ (Supplementary § S.4). Consequently every recovered root e_{31} has a bit-identical-frequency reflection twin, $e_{31}^{\text{twin}} = 2(c_{13}^E/c_{33}^E)e_{33} - e_{31} \approx 22.00 \text{ C m}^{-2}$ at this material point, and the inversion bracket used throughout ($[-8.0, -1.0] \text{ C m}^{-2}$) is chosen to exclude both the twin and the intermediate value ($e_{31} \approx 8.95 \text{ C m}^{-2}$, $\bar{e}_{31} = 0$) at which the cubic’s leading coefficient vanishes. This is the same caveat documented for the layered plate’s own open-circuit inverse [10], now derived explicitly for this operator rather than carried over by analogy alone.

5 Sensing admittance

5.1 Voltage-driven admittance is a trivial capacitor

Driving the plate electrically – imposing a bus voltage V rather than solving $Q = 0$ for it – is the natural next step after §2.3, because it is exactly this construction that produced the layered plate’s identifying resonance/antiresonance pair. On this stacking it is algebraically trivial rather than informative. The moment the driven voltage produces is identically zero for every mechanical boundary condition (§2.1), so the driven mechanical response is identically zero at every frequency that is not itself a short-circuit eigenfrequency, and the electrode charge is $Q = C_0 V$ exactly, with no motional correction at all. The undamped admittance is therefore

$$Y(\omega) = j\omega C_0 \quad (9)$$

for all ω , a frequency-independent clamped capacitor, and $k_{\text{eff}}^2 \equiv 0$ for the flexural mode of this configuration – not a computational near-zero, an algebraic identity that holds for any $\bar{c}_{11}^E$, $\bar{e}_{31}$, $\bar{\Xi}_{33}$, or geometry on this stacking. This is the direct corollary of the coincidence theorem of §2.3: a resonance/antiresonance pair requires a pole and a zero at two different frequencies, and the theorem places both at the same frequency. It is the textbook reason real bending-mode piezoelectric actuators and sensors are built as unimorphs or bimorphs and not as symmetric monolithic slabs, stated here as an exact, closed-form consequence of the radial-determinant operator rather than assumed from general electroacoustic reasoning.

5.2 Force-driven sensing admittance

Reading the same electromechanical coupling from its other port – mechanical force in, electrical charge out – is well posed, and it is not trivial. A harmonic radial ring force F at $r = r_F$ on the still-grounded, still fully electroded short-circuit free-free plate drives the two-region twelve-unknown system of §3. Its homogeneous kernel is confirmed to sit at the already-validated free-free short-circuit root: the driven system’s own determinant and the free-vibration determinant vanish at the F–F fundamental of Table 1 to within 2×10^{-13} relative, the bisection tolerance. A global force-balance identity, $2\pi \int_{r_i}^{r_o} r w(r) dr = -F/(A_2 \omega^2)$, which must hold on both the purely elastic and the piezoelectrically coupled version of this driven system at every off-resonance frequency, confirms the sign of the one inhomogeneous interface row (the shear jump) at the working precision – a check on the algebra, not a physical prediction, but one that rules out the single most consequential class of implementation error (a flipped sign on the load) independently of any charge computation.

The electrode charge integrated over the *entire* grounded face is, however, identically zero at every driven frequency, by exactly the same cancellation that produces (9), read reciprocally: the piezoelectric and dielectric contributions to the through-thickness electric displacement at the face cancel pointwise once the governing electrical equation is substituted in, leaving only an in-plane dielectric flux that the insulated-rim electrical boundary condition forces to integrate to zero over the whole face. A full-face sensing admittance would therefore be a second trivial capacitor under a different name. The nontrivial, resonance-showing output is the charge collected on a proper subset of the grounded face – a segmented electrode, both segments still shorted to ground, current read from one segment. For an inner partition $[r_i, r_*]$ of the face, Gauss’s law on the column of ceramic under the partition (insulated inner rim) makes the free charge on each face segment equal to half the radial electric flux through the cut, $Q_{\text{in}} = \pi r_* \int_{-H}^H D_r dz|_{r_*}$; the top and bottom segments carry equal charge. Kirchhoff kinematics retain only the dielectric part $\Xi_{11} E_r$ of D_r . The transverse shear stress, however, is not zero: its thickness integral is the shear resultant Q_r fixed

by equilibrium. The shear strain is then set by the constitutive law, $T_{rz} = c_{44}^E \gamma_{rz} - e_{15} E_r$, and the piezoelectric shear term $e_{15} \gamma_{rz}$ in D_r is of the same order in H as $\Xi_{11} E_r$. Eliminating γ_{rz} gives

$$\int_{-H}^H D_r dz = \frac{e_{15}}{c_{44}^E} Q_r - \frac{4H}{3} \left(\Xi_{11} + \frac{e_{15}^2}{c_{44}^E} \right) \bar{\phi}'. \quad (10)$$

At leading order $\bar{\phi}$ is proportional to $\nabla^2 w$ and Q_r to $(\nabla^2 w)'$, so both terms of (10) are proportional to $(\nabla^2 w)'$ and the segment charge is a constant multiple of its dielectric-flux part,

$$\begin{aligned} Q_{\text{in}} &= \kappa Q_{\text{in}}^{(0)}, & Q_{\text{in}}^{(0)}(r_*; \omega) &= -\frac{4\pi}{3} H \Xi_{11} r_* \bar{\phi}'(r_*; \omega), \\ \kappa &= 1 + \frac{e_{15}^2}{c_{44}^E \Xi_{11}} + \frac{e_{15}(\bar{c}_{11}^E \bar{\Xi}_{33} + \bar{e}_{31}^2)}{c_{44}^E \Xi_{11} \bar{e}_{31}}, & Y_{\text{sense}}[r_*](\omega) &:= \frac{Q_{\text{in}}(r_*; \omega)}{F}, \end{aligned} \quad (11)$$

with κ independent of ω , H , r_F , and r_* . At the material point of §4, $\kappa = -1.961$. The shear term is -3.56 times the dielectric term, opposite in sign and dominant, so the segment charge on this stacking is carried mainly by the e_{15} shear coupling, not by the dielectric flux that Kirchhoff kinematics keep. Because κ is a constant, it moves no pole or zero of Y_{sense} . It fixes the sign and the scale. Q_{in} is the free charge on the electrode segment. κ depends on c_{44}^E , which no frequency in this paper reads: at Duan's printed 73 GPa the two terms nearly cancel, $\kappa = -0.055$. It depends on the sign of e_{31} as well, through $\bar{e}_{31}$: read as $+4.1 \text{ C m}^{-2}$, e_{31} would give $\kappa = -6.258$ at 26 GPa. A measurement should therefore use measured constants.

A harmonic, undamped, axisymmetric coupled-field model tests (11) directly. It uses the elements and mesh family of §4.3, with the ring load spread over the through-thickness node line at r_F , both faces grounded, and the charge summed over the inner face nodes. It also runs at both values of c_{44}^E , because the dielectric-flux value does not depend on c_{44}^E and κ does. The ratio of the finite-element charge to $Q_{\text{in}}^{(0)}$ at $\omega = 100\text{--}1200 \text{ rad s}^{-1}$ is -1.955 to -1.960 at 26 GPa, within 0.5% of κ at every frequency. At 73 GPa, where κ is small, it is -0.0538 to -0.0550 , within 0.0022 of κ , that is within 0.2% of the dielectric-flux value the shear term almost cancels. It is mesh-converged to better than 0.01%, and the full-face charge sums to zero, as it must (Supplementary § S.7). The same decks run with $e_{31} = +4.1$ reproduce $\kappa = -6.258$ and -1.585 to within 0.5%, so the finite elements also resolve the $\bar{e}_{31}$ dependence of κ .

At the geometry of §4 with $r_F = 0.35 \text{ m}$ and a partition radius $r_* = 0.30 \text{ m}$ (chosen away from r_F so a load/cut coincidence is not built in), Y_{sense} is small and smoothly varying away from resonance and grows by three orders of magnitude, with a clean sign change, within 0.01 rad s^{-1} of the free-free short-circuit fundamental of Table 1 (Figure 3); no damping is present in this operator, so the undamped magnitude formally diverges at the pole itself and the reported sweep is a sign-change/blow-up diagnostic rather than a finite peak height. As a limiting check on the driven operator, scaling $\bar{e}_{31}$ toward zero – taking care to scale the raw material constants e_{31} and e_{33} together, since $\bar{e}_{31}$ itself is dominated by their combination at this material point and does not approach zero if only e_{31} is scaled – drives the dielectric part $Q_{\text{in}}^{(0)}$ to zero linearly, matching the mechanical Helmholtz branches to the purely elastic wavenumber over several orders of magnitude in the scaling parameter (Supplementary § S.7). The shear part of (10) does not vanish in that limit. It needs only e_{15} .

5.3 Clamped and mixed-edge sensing

The driven twelve-unknown system carries over to the other three edge pairings by the same rim-row selection as the free-vibration determinants: a clamped rim imposes $w = w' = \bar{\phi}' = 0$ in

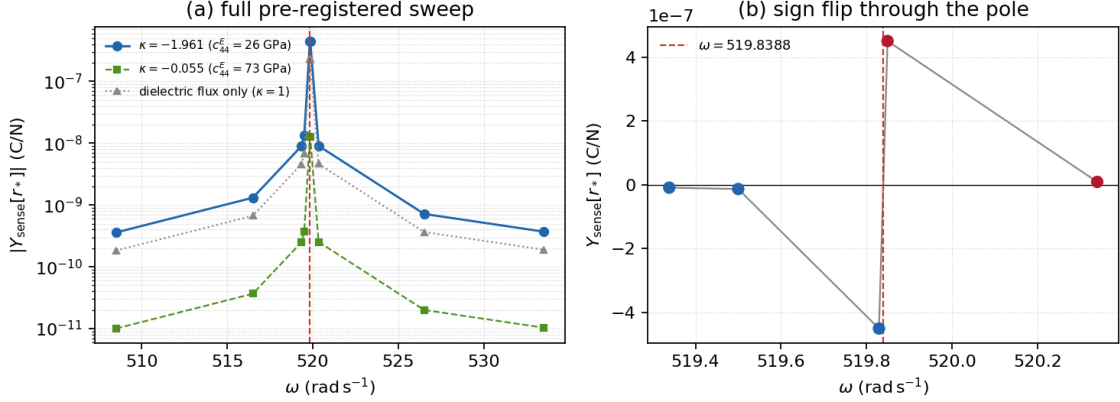


Figure 3: Segmented-electrode sensing admittance $Y_{\text{sense}}[r_*]$ (free charge per unit force on one inner face segment, (11), $c_{44}^E = 26$ GPa) near the free-free short-circuit fundamental ($519.8388 \text{ rad s}^{-1}$), $r_F = 0.35 \text{ m}$, $r_* = 0.30 \text{ m}$. (a) magnitude across the pre-registered validation sweep, with the dielectric-flux part alone ($\kappa = 1$, dotted) and Duan’s printed c_{44}^E ($\kappa = -0.055$, dashed) for comparison; (b) sign flip through the pole, undamped. Not a dense resonance curve – the nine points are the pre-registered validation gate, not a fitted or interpolated response.

place of $M_{rr} = Q_r = \bar{\phi}' = 0$, and the interface and load rows are unchanged. Its free-free path reproduces the free-free solver to 10^{-40} ; its own determinant changes sign at the clamped-clamped, clamped-free and free-clamped short-circuit roots of Table 1; a force balance that now includes the rim reactions holds to 10^{-40} at every test frequency; and Maxwell-Betti reciprocity between loads at 0.25 and 0.45 m holds to 10^{-40} . The driven κ at an interior cut is within 0.5% of the closed form at every tested frequency at 26 GPa, and within 0.004 of it at 73 GPa, where κ itself is small (Supplementary § S.7).

One thing is not a row swap. With the e_{15} term kept, the insulated rim reads $\int D_r dz = 0$ in (10), which at a free rim ($Q_r = 0$ at $n = 0$) is $\bar{\phi}' = 0$, but at a clamped rim, where the reaction Q_r is nonzero, is not. Neither $\bar{\phi}' = 0$ (the rim row used above, M1) nor the rim-corrected row with the $e_{15}Q_r$ term (M2) is a strictly valid leading-order condition inside the electrical boundary layer, about H wide, that the clamp excites. Because the segment charge depends only on D_r at the cut, a cut several H from every rim should not see the difference, and a cut close to a clamped rim will.

Twenty-four harmonic finite-element decks tested this: the four edge pairings, both c_{44}^E , and three meshes, with the clamp applied to every node of the rim line and the rims electrically insulated. The criteria were fixed before the runs. At the two interior cuts ($r_* = 0.30$ and 0.45 m , $15H$ or more from either rim) and five frequencies between 0.2 and 0.85 of each pairing’s fundamental, the finite-element segment charge agrees with $\kappa Q_{\text{in}}^{(0)}$ in all eight (pairing, c_{44}^E) cells: at 26 GPa within 2% at the four lower test frequencies and within 4% at the highest, where the dropped $O((Hk)^2)$ terms grow; at 73 GPa within 0.004 of κ in absolute terms. The dielectric-flux value alone ($\kappa = 1$) is rejected in every cell. The leading-order κ is weakest where the cut carries only inertial shear, inside the load radius next to a free rim, because the sensed charge is itself small there; a cut between the load and a clamp avoids that. Near a clamped rim ($1.25H$ and $2.5H$ from it) the finite elements are closer to M1 than to M2 at every cut, but neither model holds to 3% at both stiffnesses, which was the pre-registered condition for quoting a near-rim number. None is quoted. Cuts $15H$ or more from a clamped rim satisfy κ on every edge pairing and cuts at $2.5H$ do not; the distances in between were not tested, so the partition should sit more than about $3H$ from a clamped rim, and preferably much further.

The applied reading is the one worth stating plainly: this specimen cannot be voltage-actuated in flexure, and it cannot be sensed in flexure from a single full-face electrode either – both are exact zeros of the same underlying theorem – but it *can* be sensed, with a genuine resonant response, from a segmented electrode reading the mechanical side of the coupling. For a transducer engineer working with a symmetric, thickness-poled monolithic stack, that is an actionable design implication, not only a pair of null results: actuation on this specimen belongs on the extensional (contour) side of the same electromechanical coupling, disclosed in §4.3 and left as claim C of this series [10], while sensing of the flexural mode itself requires partitioning the electrode rather than adding instrumentation elsewhere.

6 Discussion

The layered and monolithic papers of this series now bracket the same electromechanical question from opposite stackings, using the same material, the same exact radial-determinant method, and the same inverse-recovery recipe. Offsetting a thin piezoceramic layer from a plate’s neutral axis, as in the layered construction, makes a uniform electrode voltage a flexural actuator and turns the free–free open-circuit-versus-short-circuit split into the identifying observable for e_{31} ; the short-circuit channel alone is nearly useless there, propagating 0.01% frequency precision to tens or hundreds of percent of e_{31} [10]. Removing the host and centering the same ceramic on the neutral axis, as here, removes that actuation mechanism entirely – not approximately, but as an algebraic identity holding for every mechanical boundary condition – and simultaneously turns the short-circuit channel itself into a well-conditioned observable, because the whole plate thickness is now piezoceramic rather than a thin fraction of it. Neither result is more fundamental than the other; they are the same electromechanical coupling read through two different mechanical constructions of the same material system, and a reader choosing between a layered sandwich and a monolithic ring for a real identification task should expect exactly this trade: the layered plate needs an open-circuit measurement and a rank-1 correction to identify e_{31} at all, while the monolithic ring identifies it directly from short-circuit flexure but loses the electrode-voltage resonance/ antiresonance method as a sensing route.

Limits, so they are not rediscovered as objections. No literature frequency table exists for a homogeneous, thickness-poled, Kirchhoff-class piezoceramic annulus at the time of writing; the external check here is finite-element, the same role finite elements played for the layered plate’s own open-circuit split [10], and it confirms both the coincidence theorem and the stiffening magnitude with genuinely two-dimensional coupled-field kinematics, not only within Kirchhoff theory. The coincidence theorem and the trivial voltage-driven admittance are properties of this ansatz – fully electroded faces, insulated rims – and do not extend to a plate with a partially covered or patterned electrode, which is a different electrical boundary condition and not addressed here beyond the sensing partition of §5.2. The force-driven sensing construction is axisymmetric; it holds on all four edge pairings with the partition more than about $3H$ from a clamped rim (§5.3), and no closed-form value is claimed closer to a clamp. In-plane (extensional, contour) piezoelectric coupling on this same monolithic stacking is disclosed as a genuine effect in §4.3 but is not built here; it is claim C of this series [10] and a natural companion paper on the same physical hardware, since an impedance-analyzer measurement of a real specimen would see exactly that contour response, not the flexural one studied here. A radially poled, shear-actuated specimen on Mindlin kinematics [8] is a different poling axis, a different actuation mechanism, and a different plate theory, and is not scored by this Kirchhoff monolithic operator. The sensing magnitude is a leading-order thin-plate result, confirmed by finite elements to within 0.5% on the free–free ring and 4% on clamped and mixed

edges at $r_o/H = 60$ and $c_{44}^E = 26$ GPa (to 0.004 absolute at 73 GPa, where κ nearly vanishes), and it reads c_{44}^E , e_{15} , and the sign of e_{31} , which the frequencies do not. No physical specimen is reported here; were one tested, short-circuit flexure of Table 1 is the observable for e_{31} , and a segmented-electrode measurement of Y_{sense} under mechanical excitation, not an impedance-analyzer voltage sweep, is the observable for sensing.

The elastic ring and the layered piezoelectric plate of this series [9, 10] are the baselines this construction sits on. The present frequencies do not re-score either paper’s literature tables; they extend the same exact method to a stacking neither paper addressed, and report both a coincidence theorem and a sensing alternative that the layered plate’s own geometry does not require.

7 Conclusions

A homogeneous, thickness-polarized piezoceramic ring, electroded only on its two faces, has an exact short-circuit 6×6 radial-determinant solver in the same Helmholtz/Bessel family as the corresponding layered piezoelectric plate, whose elastic limit agrees with an independent closed-ring solver to 4.8×10^{-10} relative. Because a spatially uniform electrode voltage produces a through-thickness field that is symmetric about this plate’s own neutral surface, its piezoelectric moment integrates to exactly zero: fully electroded open-circuit flexure coincides with short-circuit flexure for every mechanical boundary condition tested, not only the special case that held on the layered plate, and the voltage-driven admittance is a frequency-independent capacitor with $k_{\text{eff}}^2 \equiv 0$. Finite elements confirm both the coincidence, to eight significant figures, and the stiffening magnitude, reproducing the closed-form split to 0.03% of its value with genuinely two-dimensional coupled-field kinematics, once the electrical equation is projected consistently with the mechanical one. The piezoelectric signal on this stacking is therefore the short-circuit-versus-elastic frequency gap itself: +12.4%, +12.3%, +13.8%, and +9.6% on the free–free, clamped–clamped, clamped–free, and free–clamped $n = 0$ fundamentals of a representative PZT-4 geometry, propagating a 0.01% frequency uncertainty to 0.15%–0.17% of e_{31} – two to three orders of magnitude better conditioned than the corresponding short-circuit inverse on a layered sandwich of the same material. Because the electrode-voltage route to sensing is closed off by the same theorem that closes off actuation, sensing is read from the mechanical side instead: a harmonic radial force drives a genuinely resonant charge response on a segmented electrode, even though the same force drives an identically zero charge on the full electroded face. That segment charge is carried mainly by the transverse-shear e_{15} coupling, which Kirchhoff kinematics drop from the frequencies but which enters the charge at leading order. It is a closed-form multiple $\kappa = -1.96$ of the dielectric-flux value, and coupled-field finite elements confirm it at two shear stiffnesses; at the tabulated one, to within 0.5% on the free–free ring and 4% on the clamped and mixed edges, with the partition kept well clear of a clamped rim (more than about $3H$). A symmetric, thickness-poled monolithic plate can be neither voltage-actuated nor full-electrode-sensed in flexure, but it can be force-driven and segment-sensed – the textbook justification for unimorph and bimorph bending actuators, derived here as an exact property of the radial-determinant operator with a working sensing alternative rather than only a limitation.

Data and code availability

The solver implementing the operators of this paper, and the scripts that produced every number and table here, extend the public tree accompanying [9].

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CRedit authorship contribution statement

Garrek McCune: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization. **Daniel Stutts:** Conceptualization, Supervision.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work, the authors used generative AI tools to assist with manuscript and code organization, drafting, editing, and formatting. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of this publication.

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